

When Unknown Threat Meets Label Noise: A Self-Correcting Framework

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Abstract—Network intrusion detection systems (NIDS) are crucial for network management and security. However, in real-world scenarios, NIDS faces two core challenges: (i) label noise, where mislabeled samples in the training data distort the model’s decision boundaries; (ii) unknown attack detection, where existing methods struggle to identify novel attack patterns in dynamic attack environments. More critically, these two challenges are interlinked, forming a vicious cycle that continuously degrades the overall reliability of NIDS. Existing research often addresses these issues in isolation, and no method has yet been proposed to coordinate their antagonistic effects systematically. To tackle this open problem, we propose AEGIS-Net for the first time—a dual anti-noise framework based on multi-prototype correction and model-agnostic detection. AEGIS-Net introduces a density-difference-driven multi-prototype competition mechanism, which achieves fine-grained noise label correction through feature space sub-cluster analysis. We also design a distribution-independent k-nearest neighbors detection paradigm, using the corrected compact feature space to determine unknown attacks in open environments. The two modules are collaboratively optimized through a shared encoder, forming a positive cycle of noise suppression and detection enhancement. Extensive experiments on real-world datasets validate the effectiveness of AEGIS-Net in addressing these dual challenges. Notably, under 50% asymmetric noise conditions, AEGIS-Net achieves classification accuracy of 89.02% for known attacks and 98.76% for unknown attack detection on the MAL_TLS2023 dataset. Theoretical proofs and visualization analysis reveal the anti-noise properties of AEGIS-Net under feature space stability constraints.

Index Terms—Network intrusion detection system, unknown attack, encrypted traffic.

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Our code is available at <https://github.com/niebiakong/AEGIS-Net>.

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I. INTRODUCTION

CURRENT malicious traffic detection methods, especially the intelligent methods that have been widely used in recent years, heavily rely on prior knowledge [1], [2], [3]. As attackers’ behaviors become increasingly covert and new attack patterns continue to emerge, obtaining precise prior knowledge becomes more challenging. Specifically, attackers use techniques such as traffic shaping and encrypted payloads to make malicious traffic highly overlap with normal traffic in terms of statistical features [4], [5], [6]. Manual or automated labeling tools struggle to differentiate such obfuscated traffic, leading to the mislabeling of malicious samples as normal or normal traffic as attacks, thereby introducing label noise in the dataset.

We summarize the main manifestations of weak prior knowledge in network intrusion detection (NIDS) into two typical scenarios: the first is label noise, where some malicious traffic in the dataset is incorrectly labeled as normal or as other malicious behaviors; the second is unknown attacks, where the model encounters certain attack categories that were not present in the training dataset after deployment. Label noise significantly weakens the model’s generalization performance by causing feature space distribution shifts and decision boundary distortion [7], [8], [9]. Experiments show that even in scenarios with low noise rates ($\leq 5\%$), the upper limit of classification accuracy is still suppressed at 89% [10]. The model, lacking labels for unknown attacks, is unable to recognize such traffic, leading to a sharp increase in false positive rates [11]. Furthermore, if NIDS includes these misclassified samples in the training set, it results in new label noise. This misclassification continuously pollutes the training set, forming a vicious cycle of “noise-false detection-more noise,” causing the overall reliability of NIDS to decline.

In real-world scenarios, the challenges of label noise and unknown attacks, two major weak prior knowledge issues, co-exist, leading to a sharp decline in the detection performance of malicious behaviors in NIDS. Therefore, NIDS faces two difficult tasks: T1: *the accurate classification of known malicious attacks*, and T2: *the precise detection of unknown attacks*.

For T1, existing methods assume that the training data labels are sufficiently accurate and cover the known attack categories. This assumption is actually difficult to satisfy, as automated labeling tools may produce mislabels due to protocol parsing errors or feature extraction biases. In contrast, manual labeling is easily limited by the experience level of security analysts [12].

Especially in the context of the widespread use of encryption protocols and the upgrading of traffic obfuscation technologies [13], the reliability of traditional rule-based or signature-based labeling methods continues to decrease. For T2, existing methods need to construct a highly discriminative latent feature space to distinguish between known and unknown categories and also design robust confidence evaluation mechanisms to reject uncertain samples [14]. Under the premise of label noise, these current methods do not have effective countermeasures.

Current NIDS are facing the coupling effect of label noise and unknown attacks, which fundamentally stems from the dependency on the closed-world assumption in supervised learning and the unreliability of prior knowledge in a dynamic attack environment. Therefore, the first challenge of our work is **CH1**: *How to design methods for handling label noise in weak prior knowledge?* The second challenge of our work is **CH2**: *How to design noise-resistant detection methods for unknown malicious traffic?*

There are two common strategies used to address **CH1**: loss correction [15], [16] and sample selection [17], [18]. Loss correction methods directly modify the design of the loss function or the assumption of the label distribution, allowing the model still to learn the correct feature mapping despite label noise. This method heavily depends on the accurate estimation of the noise distribution, such as the noise transfer matrix. Sample selection methods dynamically identify potential clean samples, using only high-confidence samples for training or assigning weights to samples to reduce the impact of noise [19]. Sample selection can explicitly exclude simple noisy samples, but it is prone to discarding difficult clean samples, leading to a fuzzy decision boundary for the model. In addressing **CH2**, whether using discriminative or generative models, the core challenge lies in managing uncertainty in open environments [14], [20]. When facing the complex scenario of noise-unknown attack coupling, theoretical shortcomings still exist.

To address **CH1**, we observe that there is a density difference between clean samples and noisy samples. Additionally, since network traffic in high-dimensional representation spaces often contains multiple sub-clusters, we select multiple prototypes for each category of sample to infer the true label of noisy samples, achieving precise classification of known malicious traffic. To address **CH2**, we do not make any assumptions about the underlying distribution of traffic, and model-independent, based on the distance of the sample to the known class, we distinguish between known and unknown class traffic.

Considering both **CH1** and **CH2**, we propose a novel intrusion detection system, AEGIS-Net, under the scenario of weak prior knowledge. Its innovation lies in two aspects: First, a multi-prototype dynamic correction mechanism based on density heterogeneity is constructed. By exploring the multi-subcluster distribution characteristics of known class traffic, a dynamic prototype set is established to accurately locate noise samples and correct their labels, overcoming the limitations of single-prototype representation in modeling intra-class diversity. Second, a non-parametric detection paradigm based on the relative distance ratio of K-nearest neighbors is designed. An adaptive threshold mechanism constructs a distribution-independent open space metric system, eliminating

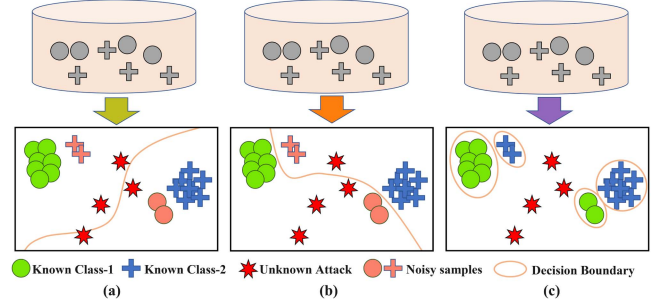


Fig. 1. Early methods considered CH1 and CH2 in isolation. (a) Decision boundary of a closed-world classifier under noisy labels in an open environment. (b) Decision boundary of a closed-world classifier under noisy label learning in an open environment. (c) Decision boundary of AEGIS-Net in an open environment.

reliance on traffic distribution assumptions. The two modules form a bidirectional enhancement mechanism through a shared feature encoding network: the label correction process enhances the feature compactness of known classes. At the same time, the model-independent k-nearest neighbor detection leverages the corrected feature space to improve the separability of unknown attacks, forming a positive feedback loop of noise suppression and detection enhancement. We prove both theoretically and experimentally that the detection performance of AEGIS-Net is independent of the noise rate under the premise of feature space stability. A comparison between our AEGIS-Net and existing methods is visualized in Fig. 1.

To the best of our knowledge, our work is the first intrusion detection model to comprehensively consider both label noise and unknown attack detection in a weak prior knowledge scenario. In summary, this paper provides the following contributions:

- To address **CH1**, we propose a density difference-driven multi-prototype label noise correction mechanism, achieving fine-grained label correction through feature space sub-clusters.
- To address **CH2**, we design a model-agnostic paradigm for unknown traffic detection, eliminating the reliance on specific distribution assumptions found in traditional methods. We theoretically prove that AEGIS-Net exhibits noise-resistant properties in open-world scenarios.
- We validated the effectiveness and efficiency of AEGIS-Net on four public datasets for both **CH1** and **CH2**. Compared to the current SOTA, AEGIS-Net achieves optimal performance. Even in high-noise scenarios, the performance of unknown malicious traffic detection remains unaffected by noise rates.

The remainder of this paper is structured as follows: Section II presents an overview of the related work, followed by a problem statement in Section III. Subsequently, the method design is detailed in Section IV. The proof of theory is given in Section V. Moreover, our approach is evaluated in Section VI. Finally, we conclude the paper in Section VIII.

II. RELATED WORK

In this paper, we only review various classification models related to *noise label handling* and *unknown malicious traffic detection*. In Table I, we give a summary for each related work.

TABLE I
SUMMARY OF RELATED WORK

	Year [ref]	Feature Extraction	Algorithm	Method Type	Noisy Label	Open-World	Code
Noise Label Handling	2015 [21]	Statistical Features	Random Forest	Traditional ML	✓	✗	✗
	2017 [22]	Statistical Features	Random Forest	Traditional ML	✓	✗	✗
	2018 [23]	Figures	CNN	Deep Learning	✓	✗	✓
	2023 [19]	Statistical Features	CNN, k-means	Deep Learning	✓	✗	✓
	2023 [24]	Sequence Features	AE, GRU, GAN	Deep Learning	✓	✗	✓
	2023 [25]	Statistical Features	DRO	Deep Learning	✓	✗	✗
	2025 [26]	Statistical Features	Instance Relabeling	Deep Learning	✓	✗	✓
	2025 [27]	Statistical Features	CNN, Attention	Deep Learning	✓	✗	✗
Unknown Traffic Detection	2020 [28]	Traffic Image	CNN	Deep Learning	✗	✓	✗
	2021 [29]	Statistical Features	k-means	Traditional ML	✗	✓	✓
	2021 [30]	Payload Features	Siamese network	Deep Learning	✗	✓	✗
	2021 [31]	Statistical Features	VAE, EVT	Deep Learning	✗	✓	✓
	2022 [32]	Statistical Features	GAN	Deep Learning	✗	✓	✗
	2023 [33]	Statistical Features	One-class classifiers	Deep Learning	✗	✓	✗
	2023 [34]	Statistical Features	GAN	Deep Learning	✗	✓	✗
	2023 [35]	Statistical Features	CFSFDP [36]	Traditional ML	✗	✓	✗
	2023 [37]	MAV	k-Logit neighbor	Deep Learning	✗	✓	✓
	2023 [38]	TAM	CNN	Deep Learning	✗	✓	✓
	2024 [39]	Statistical Features	CNN	Deep Learning	✗	✓	✓
	2024 [40]	Statistical Features	Cost-sensitive matrix	Deep Learning	✗	✓	✗
	2024 [41]	Statistical Features	AE	Deep Learning	✗	✓	✓
	2024 [42]	Traffic Image	Multi-binary classifier	Deep Learning	✗	✓	✗
AEGIS-Net		Statistical Features	CNN, KNN	Deep Learning	✓	✓	✓

AE: Autoencoders, CNN: Convolutional Neural Networks, GAN: Generative Adversarial Networks, VAE: Variational Autoencoders, EVT: Extreme Value Theory, DRO: Distributionally Robust Optimization, MAV: Mean Activation Vector, Traffic Image: Grayscale Image, TAM: Traffic Aggregation Matrix.

A. Noise Label Handling

Noise label handling methods can be summarized into Loss Correction, Sample Selection, Semi-Supervised Learning, and Meta-Learning, among others. MCRE [19] balances robust training and data cleaning, resulting in a cleaner dataset and a more accurate classification model. UTC [21] employs a consensus filtering mechanism with an ensemble of classifiers, using a veto strategy to identify and remove strong noise samples. The introduction of Tolerant Weight adjusts the contribution of suspected noisy samples in training, reducing the noise impact. NSTC [22] enhances traffic classification performance under noisy environments by integrating multiple classifiers to detect noise, weighted sample selection, and combining robust classifiers. Co-teaching [23] uses two neural networks, with the current network passing the samples it considers clean to the other network for training, thereby avoiding noise accumulation. DSDIR [26] proposes a distribution-aware sample selection method to re-label samples while addressing both noise label and long-tail issues. TRACE [27] utilizes a multi-loss optimization mechanism to jointly optimize feature completeness, intra-class compactness, and inter-class distinguishability, generating noise-resistant traffic representations.

B. Unknown Malicious Traffic Detection

CBSeq [20] employs channel-level behavior clusters as detection units, extracting four types of behavior sequences and using independent sub-networks for each. The self-attention

mechanism captures internal relationships within sequences to distinguish malicious from benign traffic. In the work by Qing et al. [24], they leverage neural networks to synthesize unknown traffic, effectively converting the open-world problem into a closed-world classification challenge. The work in [25] introduces Wasserstein DRO into federated learning, leveraging noisy data by normalizing it into a robust region defined by the Wasserstein-2 distance, dynamically adjusting the distribution of noisy data to align with clean data. RoFi [38] develops a robust traffic representation method that generates a Traffic Aggregation Matrix (TAM) using packet counts and time series to fully capture key information features leaked in the traffic. RFG-HELAD [39] integrates multiple machine learning/deep learning models into a single framework using heterogeneous ensemble learning, addressing the coexistence of unknown attacks and adversarial attacks in fine-grained attack detection scenarios. Yang et al. [43] proposes a lightweight method for detecting unknown malicious traffic by utilizing a variational autoencoder and EVT theory to model reconstruction errors, with dynamic model updates through model distillation. The authors in [44] presents RAGN (Robust Adaptive Graph Neural Network) for detecting unknown malicious traffic and improving resilience to adversarial attacks, along with the SPAN (Semantic-Preserving Adversarial Node Injection) attack method to evaluate defense capabilities in dynamic environments. HyperVision [45] models IP addresses as nodes and traffic patterns as edges, reducing graph density through flow aggregation and feature recording and employing unsupervised graph learning techniques like

connectivity analysis and key vertex clustering to detect anomalous interactions. Kitsune [46] continuously learns the pattern of normal traffic through an autoencoder. Once behavior deviates from this normal pattern, it is flagged as anomalous based on the reconstruction error. However, if attack traffic is mixed into the deployment environment from the beginning, the model will incorrectly learn the abnormal behavior and will no longer classify it as anomalous in the future. This is a limitation of Kitsune.

C. In Summary

Existing research has made significant progress in addressing noisy labels and unknown malicious traffic detection. However, these two critical challenges are often tackled in isolation, leading to suboptimal performance in real-world scenarios where both issues coexist. To address the above limitations, this paper proposes a collaborative optimization framework for noise robustness and unknown attack detection. Through dynamic label correction and a flexible, generalizable, non-parametric open-world detection paradigm, it achieves accurate detection of unknown malicious traffic in noisy environments.

III. PROBLEM STATEMENT

The problem of simultaneously solving **CH1** and **CH2** can be defined as follows:

Let $\tilde{V} = \{x_i, y_i\}_{i=1}^N$ be the training data of known classes traffic (possibly containing noise), and notate R_O and R_ϵ represent the unknown risk and empirical risk, respectively. The goal of detecting unknown malicious traffic is to find a measurable function $f \in \mathcal{H}$, where $f(x) > 0$ indicates correct identification. The following formula determines the function f :

$$\arg \min_{f \in \mathcal{H}} \left(R_O(f) + \lambda_r R_\epsilon \left(f \left(\tilde{V} \right) \right) \right), \quad (1)$$

where λ_r is a regularization constant.

From observing this function, we can derive the following conclusions:

There is a trade-off between empirical risk and unknown risk. When the empirical risk is large, it indicates poor model fitting on the training set, and the closed-world classification performance is suboptimal. In this case, if the model fails to effectively distinguish the known classes, its decision boundary may not be clear enough, making it more prone to misclassifying unknown classes, thereby increasing the unknown risk and degrading the performance of unknown class detection. When the empirical risk is small, although the closed-world classification performance is better, overfitting may occur, resulting in decreased generalization ability of the model. In this case, the model may overly rely on the distribution features of the training data, reducing its robustness to unknown classes, which may actually increase the unknown risk. The goal of (1) is to adjust the importance of both risks by the weight λ_r , but the selection of λ_r during the optimization process is challenging.

In the case of having only known-class traffic, the unknown risk R_O cannot be directly computed or optimized because R_O

TABLE II
SYMBOLS AND DEFINITIONS USED IN THIS PAPER

Symbols	Definitions
$\tilde{V} = \{x_i, y_i\}_{i=1}^N$	The training dataset
$E(\cdot)$	The encoder of AEGIS-Net
$h(\cdot)$	The fully connected layer of AEGIS-Net
L_{Cla}	The classification loss
$L_{Contrast}$	The contrastive loss
L_{total}	The total loss
α, β	The weight factors in L_{total}
ρ	The density of traffic sample
η	The similarity metric of traffic sample
$NN_k(z_{test})$	The distance to the k -th nearest neighbor of z_{test}

involves the integration over the open space, which cannot be computed explicitly.

When there are noisy labels in the training set, the noisy labels introduce incorrect supervisory signals, causing the model to learn erroneous feature associations. In this case, the model may fail to converge effectively, and the empirical risk will significantly increase. Suppose the model forcibly fits noisy labels, such as memory noise samples. In that case, it may result in the empirical risk being underestimated on the training set, but the true generalization ability decreases, and the closed-world classification remains worse. Additionally, noisy labels distort the distribution of the known classes, leading the model to learn incorrect decision boundaries, making it more likely to misclassify unknown class traffic as known classes, which increases the unknown risk. As the noise rate increases, the closed-world classification performance continues to decline, and the empirical risk increases overall, which also increases the unknown risk, resulting in worse unknown class detection performance.

To minimize the interference of noisy labels, a direct approach is to detect the noisy samples in the training dataset $\tilde{V}$ and correct the labels of the noisy samples, thereby reducing their impact on the function as much as possible.

Formally, given a training set $\tilde{V}$, where x is a traffic sample and y is its annotated label with $y \in L = \{1, 2, \dots, c\}$. We denote the true label of x_i as y_i^* . Conventionally, we assume that all annotated labels of the traffic samples are correct, i.e., $y_i = y_i^*$. Thus, we can use the cross-entropy loss function to optimize the model parameters and achieve an accurate classification of the traffic. However, the assumption of clean labels is too strict for real-world scenarios, and noisy labels are unavoidable in real-world datasets. Therefore, our goal is to design a noise-resistant method for unknown malicious traffic detection. The symbols and definitions used in this paper are shown in Table II.

IV. OUR APPROACH

Our system is designed based on a dual strategy: First, by analyzing the clustering characteristics of encrypted traffic in a high-dimensional feature space, we dynamically select multiple representative samples as reference benchmarks for each known traffic class. By measuring the similarity differences between samples, we identify and correct mislabeled traffic data, thereby

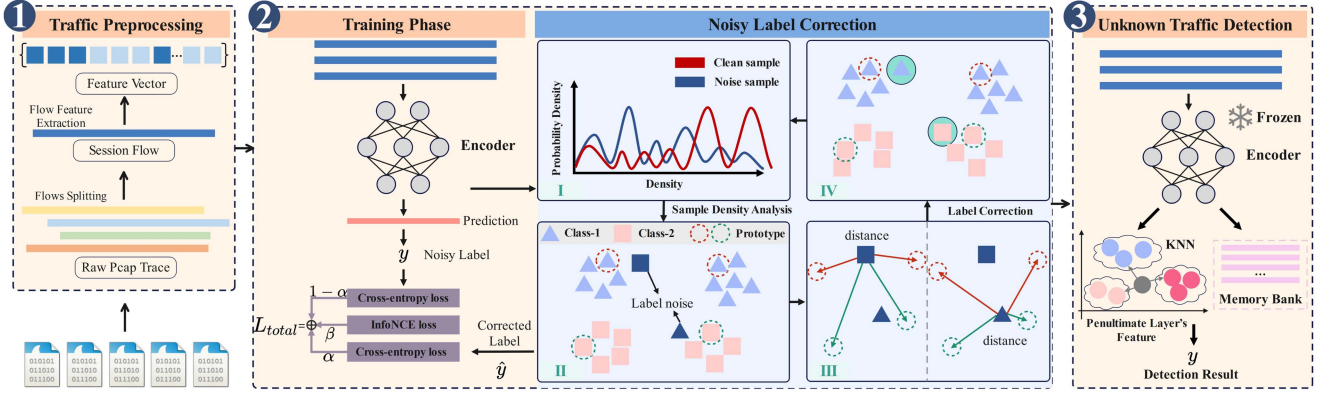


Fig. 2. The overview of AEGIS-Net. After the initial training is completed using noisy labels during the training phase, it proceeds to the noisy label correction phase. The noise label correction phase consists of four steps: I: Calculate the density and similarity metric for each sample, II: Prototype selection, III: Voting decision, IV: Noise sample label correction. These four steps are alternated and repeated in each epoch.

alleviating the interference of noisy labels on model learning. Second, during the unknown attack detection phase, we employ a model-independent mechanism based on neighboring sample distance comparison. By calculating the distance between the test sample and known categories, we determine whether it belongs to an unknown attack type, avoiding any pre-specified assumptions about the data distribution. Both modules share a feature encoding network, which enhances the intra-class cohesion of the feature space through noise correction, while the optimized compact features improve the ability to distinguish unknown classes, forming a closed-loop optimization process.

The overall framework of AEGIS-Net is shown in Fig. 2, which consists of three modules: the traffic preprocessing module, the model training module, and the unknown malicious traffic detection module. The traffic preprocessing module extracts statistical features from the raw network flows and transforms them into feature vectors. The model training module corrects the labels of the training data through noise label correction, thereby obtaining a reliable network representation to assist downstream tasks. Based on this, the model has learned the correct decision boundaries. The unknown malicious traffic detection module uses the KNN algorithm to detect unknown malicious traffic.

A. Traffic Preprocessing

The traffic preprocessing module transforms raw network traffic into a one-dimensional feature vector. As shown in Fig. 2, we divide the raw network traffic into bi-directional flows according to five tuples. We use CICFlowMeter¹ to compute the features of each flow. In the experiments, we remove categorical features such as Flow ID, Src IP, Src Port, Dst IP, Dst Port, and Timestamp, keeping only numerical features to prevent the model from memorizing specific information and instead focus on learning the inherent patterns of attacks.

B. Training Phase

The core objective of the training phase is to optimize the parameters θ of the Convolutional Neural Network (CNN) so that the feature extractor E acquires discriminative representation capabilities. Standard training uses the cross-entropy loss function:

$$L(F(\theta, x), y) = -\frac{1}{n} \sum_{i=1}^n y_i \log F(x_i, \theta), \quad (2)$$

where n is the mini-batch size, and y_i is the label corresponding to the feature vector x_i . Since real-world data may contain label noise (i.e., y_i could be incorrect), a dynamic correction mechanism is introduced to generate corrected labels $\hat{y}$. Based on the original and corrected labels, a robust classification loss function is constructed:

$$L_{Cla} = (1 - \alpha)L(F(x, \theta), y) + \alpha L(F(x, \theta), \hat{y}). \quad (3)$$

To enhance the structure of the feature space, a contrastive learning mechanism is further introduced. The InfoNCE loss function [47] is used to improve intra-class compactness and inter-class separability:

$$L_{Contrast} = -\frac{1}{n} \sum_{i=1}^n \log \left(\frac{\exp(q_i \cdot k_{i+}/\tau)}{\sum_{j=1}^n \exp(q_i \cdot k_{j-}/\tau)} \right), \quad (4)$$

where q_i is the encoding vector of the query sample i , k_{i+} is the encoding vector of the positive sample corresponding to query sample i , k_{j-} is the encoding vector of a negative sample not corresponding to query sample i , and τ is the temperature parameter that controls the distribution of similarity scores.

Considering the above objectives, the overall loss function is defined as:

$$\begin{aligned} L_{total} &= L_{Cla} + \beta L_{Contrast} \\ &= (1 - \alpha)L(F(x, \theta), y) + \alpha L(F(x, \theta), \hat{y}) + \beta L_{Contrast}, \end{aligned} \quad (5)$$

where $\alpha, \beta \in [0, 1]$ are balancing weights. The training process is divided into two stages: 1) *Initial Pretraining*: For the first

¹<https://github.com/ahlashkari/CICFlowMeter>

N_{pre} epochs, set $\alpha = 0$, and only use the original noisy labels y to train the model. 2) *Co-Optimization*: Once the model has acquired preliminary representation capabilities, the Noisy Label Correction module is activated to generate $\hat{y}$, and the parameters θ are jointly optimized using the dual loss.

In the Noisy Label Correction module, our goal is to obtain the correct label for each network flow to provide supervisory information for model training. For label correction, the first step is to select several class prototypes for each type of traffic. Inspired by the CFSFDP [36] algorithm, we first extract deep features of the network flows using the preliminary network trained in the first stage. In the experiment, the model consists of a feature extraction network E and a classification network h , so $F(x, \theta) = h \cdot E(x)$, where h is a fully connected layer.

To select appropriate class prototypes for the c -th class, we first extract a set of deep features $\{E(x_i)\}_{i=1}^n$ from the traffic samples $\{x_i\}_{i=1}^n$ with the same noisy label c , where n is the number of samples with the noisy label c . We then use cosine similarity to calculate the similarity between the deep features of the c -th class samples, constructing a similarity matrix $S \in \mathbb{R}^{n \times n}$, where $S_{ij} = \cos(E(x_i), E(x_j)) \in [-1, 1]$. S_{ij} represents the similarity between any two samples x_i and x_j . The larger S_{ij} is, the more similar the two samples are. However, the problem arises that as n increases, the computation and storage of the matrix S will become extremely time-consuming and expensive. Therefore, here, we only randomly sample m samples from the n samples, with $m < n$, and thus only need to compute the matrix $S^{m \times m}$ to reduce the computational cost.

To select appropriate prototypes, we define a density ρ_i for each sample x_i based on the similarity matrix S :

$$\rho_i = \sum_{j=1}^m \chi(S_{ij} - S_c) \in (-m, m), \quad (6)$$

where S_c is the truncation density, and $\chi(\cdot)$ is the sign function. S_c is a constant, which is the value of the element in the matrix S after sorting all elements from small to large and selecting the top 40%. Experiments show that the value of S_c has no significant impact on the experimental results.

From the above definition of density ρ , it can be observed that the larger the density of a sample, the more its surrounding samples are. Samples with correct labels should be close to each other, while samples with noisy labels usually form clusters that are far from their corresponding noisy label samples. In Fig. 3(a), the density of noisy label samples and correctly labeled samples is shown. It can be seen that noisy label samples generally have a smaller density, while correctly labeled samples tend to have a larger density. In other words, samples with higher density are often clean samples from which we can select prototypes for the class. If p prototypes are needed, the samples with the top p largest density values are considered class prototypes. However, this introduces a problem: if these p prototypes are all in the same sub-cluster, these p prototypes are essentially the same prototype and do not serve the purpose of representing multiple sub-clusters within a class.

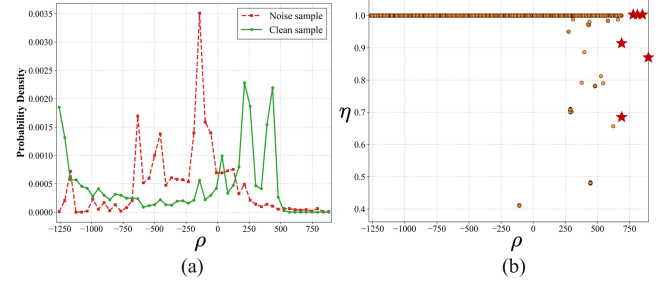


Fig. 3. (a) Density ρ of noisy and clean samples in the MaL_TLS2023 dataset. (b) Distribution of the density ρ and similarity metric η . Red stars indicate the samples with the top-6 highest ρ values.

Therefore, we define a similarity metric for each sample:

$$\eta_i = \begin{cases} \max_{j, \rho_j > \rho_i} S_{ij}, & \rho_i < \rho_{\max} \\ \min_j S_{ij}, & \rho_i = \rho_{\max} \end{cases}, \quad (7)$$

where $\rho_{\max} = \max\{\rho_1, \rho_2, \dots, \rho_m\}$. From this definition, it can be seen that for the sample x_i with the largest density, the similarity metric η_i is the smallest; for the other low-density samples x_i , their similarity metric η_i is defined as the similarity between x_i and x_j with a higher density value than x_i . From the above definition, it can be seen that a smaller similarity metric η_i indicates that the features of sample x_i are not closely related to other high-density samples. Therefore, samples with a high-density value ρ (higher likelihood of clean labels) and a low similarity metric η (clean label, but moderately far from other clean labels) meet our criteria for selecting class prototypes.

In Fig. 3(b), the red stars are the samples with the largest density ρ . Samples with high density ρ tend to have relatively small similarity metric η . In this paper, we select prototypes with $\eta < 0.95$ from the top-ranked samples. Even if the number of prototypes that meet the conditions after filtering is less than p , we will still select the p smallest indices based on the similarity metric η in the engineering implementation to ensure that the prototype set remains complete. In other words, regardless of the density ρ and the similarity measure η , each category will ultimately have p prototypes.

After selecting the prototype sets

$$\{E(x_{i1, i2})\}_{i1=1,2,\dots,c; i2=1,2,\dots,p}, \quad (8)$$

for each known class, where c is the number of known traffic classes, and $\{E(x_{i1}), E(x_{i2}), \dots, E(x_{ip})\}$ represents the prototype set for the i -th known traffic class. Given an unlabeled sample x , we first obtain its embedding representation $E(x)$, and then calculate the similarity between $E(x)$ and the prototype sets of different classes:

$$\sigma_c = \frac{1}{p} \sum_{l=1}^p \cos(E(x), E(x_{il})), \quad i = 1, 2, \dots, c. \quad (9)$$

Here, we use the average similarity over the p prototypes. This voting decision method prevents the misclassification of some hard samples that have nearly identical high similarities into different classes. We can then obtain the corrected label $\hat{y} \in$

Algorithm 1: The Training Process of AEGIS-Net.

Input: Noisy labeled dataset $\tilde{V}$, hyperparameters α, β, p, m

Output: Trained model $F = h \cdot E$ with robust feature representation

- 1: Initialize neural network parameters θ for E and h
- 2: Split $\tilde{V}$ into mini-batches $\{B_1, B_2, \dots, B_T\}$
- 3: **for** epoch = 1 to N_{total} **do**
- 4: **if** epoch $\leq N_{pre}$ **then** $\triangleright$ Pre-training with noisy labels
- 5: **for** each mini-batch B_t **do**
- 6: Calculate classification loss L_{pre} using (5), where $\alpha = 0$.
- 7: $\theta^{t+1} \leftarrow \theta^t - \xi \nabla_{\theta^t} L_{pre}$
- 8: **end for**
- 9: **else** $\triangleright$ Noise label correction
- 10: **for** each class $c \in \{1, \dots, c\}$ **do**
- 11: Extract deep features $\{E(x_i)\}$ for samples with noisy label c .
- 12: Randomly sample m features $\{z_1, \dots, z_m\}$.
- 13: Calculate similarity matrix $S \in \mathbb{R}^{m \times m}$.
- 14: Calculate density ρ_i using (6).
- 15: Calculate similarity metric η_i using (7).
- 16: Select p prototypes.
- 17: **end for**
- 18: **for** each mini-batch B_t **do**
- 19: Obtain corrected labels $\hat{y}$ using (10).
- 20: Calculate classification loss L_{Cla} using (3).
- 21: Calculate contrastive loss $L_{Contrast}$ using (4).
- 22: Total loss $L_{total} = L_{Cla} + \beta L_{Contrast}$.
- 23: $\theta^{t+1} \leftarrow \theta^t - \xi \nabla_{\theta^t} L_{total}$
- 24: **end for**
- 25: **end if**
- 26: **end for**

$\{1, 2, \dots, c\}$, where

$$\hat{y} = \arg \max_i \sigma_i, \quad i = 1, 2, \dots, c. \quad (10)$$

After obtaining the corrected label data, we use it as supplementary supervisory signals to train the neural network F during the training phase. The training process of our model is shown in Algorithm 1.

C. Unknown Traffic Detection

After completing the training process, the model F has corrected noisy labels and learned deep representations of traffic, achieving intra-class compactness and inter-class separation in the representation space. Many unknown malicious traffic detection methods constrain each known class to a specific distribution in the representation space [31], [43], but this approach is difficult to achieve. Our method makes no assumptions about the underlying feature space and detects unknown malicious traffic based on distance or density.

The KNN-based detection method calculates the distance between each test sample and training sample in the embedding

Algorithm 2: Unknown Malicious Traffic Detection.

Input: Test sample x_{test} , training features $\Theta = \{z_1, \dots, z_n\}, k, \lambda$

Output: Detection result $D \in \{\text{Known}, \text{Unknown}\}$

- 1: Extract penultimate layer feature: $z_{test} = E(x_{test})$.
- 2: Normalize feature: $z_{test} \leftarrow z_{test} / \|z_{test}\|_2$.
- 3: Calculate distance set $\Delta \leftarrow \{\|z_{test} - z_i\|_2 \mid z_i \in \Theta\}$.
- 4: Sort Δ in ascending order to get $\{d_{(1)}, \dots, d_{(n)}\}$.
- 5: Retrieve k -th smallest distance: $d_{(k)} \leftarrow \Delta[k]$.
- 6: **if** $d_{(k)} < \lambda$ **then**
- 7: $D \leftarrow \text{Known}$
- 8: **else**
- 9: $D \leftarrow \text{Unknown}$
- 10: **end if**
- 11: **return** D

space. This decision-making method can be represented by the following formula:

$$D(x_{test}) = \begin{cases} \text{Known}, & \text{if } S(x_{test}) \leq \lambda \\ \text{Unknown}, & \text{if } S(x_{test}) > \lambda \end{cases}, \quad (11)$$

where $S(x)$ is the score function, and a larger value of $S(x)$ indicates a higher probability that the test sample is classified as an unknown class. λ is the threshold.

Specifically, we use the normalized penultimate layer features $z = E(x) / \|E(x)\|$ for detecting unknown malicious traffic. Assuming the training set is represented in the embedding space as $\Theta = \{z_1, z_2, \dots, z_n\}$. During the detection phase, the normalized representation z_{test} is first calculated. Then, for each training sample in Θ , the distance $d_i = \|z_{test} - z_i\|$ is computed, where $1 \leq i \leq n$. The distance values d_i are then arranged in ascending order to form the reordered distance vector $\Delta = \{d_{(1)}, d_{(2)}, \dots, d_{(n)}\}$, where $\forall i < j, d_{(i)} < d_{(j)}$. The decision function can be represented as:

$$S(z_{test}, k) = \begin{cases} \text{Known}, & \text{if } NN_k(z_{test}) \geq \lambda \\ \text{Unknown}, & \text{if } NN_k(z_{test}) < \lambda \end{cases}, \quad (12)$$

where $NN_k(z_{test}) = d_{(k)}$ represents the distance to the k -th nearest neighbor of z_{test} . The threshold λ is usually chosen to ensure a high proportion of known class samples (e.g., 95%) are correctly classified. At this point, the threshold λ does not depend on the unknown malicious traffic class because unknown malicious traffic is invisible. We summarize our unknown malicious traffic method in Algorithm 2.

The advantage of KNN is its simplicity and ease of understanding, and it does not require training. However, its disadvantage is the high computational cost, especially on large datasets, as it requires calculating the distance between the test sample and all samples in the dataset. We choose to use the Approximate Nearest Neighbor (ANN) algorithm for vector retrieval [48]. Unlike KNN, ANN does not always return the exact nearest neighbors, but returns approximate nearest neighbors within a certain error margin, trading off retrieval speed for accuracy.

V. THEORETICAL JUSTIFICATION

In this section, we provide proof that AEGIS-Net uses KNN for unknown malicious traffic detection and that the results of unknown malicious traffic detection are independent of noise.

Definition 1. (Feature Space Stability): Let the feature extractor be E^* in the noise-free case, and E be the feature extractor obtained after label correction and contrastive learning in the noisy case. The feature distribution shift is defined as:

$$\Delta_E = E_{x \sim D} [\|E(x) - E^*(x)\|_2], \quad (13)$$

where D is the distribution of the known class training sample. It is assumed that after label correction and contrastive loss optimization, $\Delta_E < \varepsilon$, where ε is a small constant.

Definition 2. (Non-parametric Detection Mechanism): The unknown malicious traffic detection phase uses the KNN algorithm. For a test sample x_{test} , define its KNN distance to the training set $\tilde{V}$ as $NN_k(x_{test}) = d_{(k)}$. The threshold λ is determined by the γ -quantile of the known class distance distribution:

$$\lambda = \inf \{t \mid P_{x \sim D} (d_{(k)}(x) \leq t) \geq \gamma\}. \quad (14)$$

Lemma 1. (Proof in Appendix-A): According to the Feature Space Stability assumption, for any sample x , $\|E(x) - E^*(x)\|_2 \leq \varepsilon$. Therefore, for any two samples x_i and x_j , their feature distance shift satisfies:

$$\|E(x_i) - E(x_j)\|_2 + \|E^*(x_i) - E^*(x_j)\|_2 \leq 2\varepsilon. \quad (15)$$

Lemma 2. (Proof in Appendix-B): For the test sample x_{test} , the difference in its KNN distance between the noisy and noise-free cases satisfies:

$$|d'_{(k)}(x_{test}) - d_{(k)}(x_{test})| \leq 2\varepsilon. \quad (16)$$

Theorem 1. (Proof in Appendix-C): Given that the γ -quantile λ of the class distance distribution satisfies $P(d_{(k)}(x) \leq \lambda) \geq \gamma$ when the feature distribution shift $\Delta_E < \varepsilon$, the cumulative distribution function (CDF) shift of the distance distribution satisfies:

$$|F_r(t) - F_0(t)| \leq \frac{2\varepsilon}{\sigma}, \quad (17)$$

where σ is the standard deviation of the distance distribution (assuming $\sigma \gg \varepsilon$). Therefore, the change in threshold satisfies:

$$|\lambda(r) - \lambda(0)| \leq 2\varepsilon, \quad (18)$$

where $r \in [0, 1]$ is the noise rate.

The premise of Theorem 1 is the feature space stability, i.e., $\Delta_E < \varepsilon$, where ε is a small constant. In experiments on public datasets, we computed the mean value of Δ_E as 1.210 with a standard deviation of $\text{std} = 0.06$ on the Mal_TLS2023 dataset. This shows that AEGIS-Net satisfies the feature space stability. Furthermore, the conclusion of Theorem 1 is that the threshold variation $|\lambda(r) - \lambda(0)| \leq 2\varepsilon$. Fig. 4 shows the detection thresholds of AEGIS-Net on different datasets. In the case of symmetric noise, $\lambda(0)$ on the Mal_TLS2023 dataset is 0.0099, $\lambda(0.1) = 0.0264$, and $\lambda(0.5) = 0.0527$, which satisfies the conclusion of Theorem 1.

TABLE III
EXPERIMENTAL SETUP FOR KNOWN AND UNKNOWN CLASSES OF TRAFFIC IN PUBLIC DATASETS

Characteristics	Mal_TLS2023	CICIDS2017	CICIDS2018
# Attack types	23	14	3
# Normal samples	0	2272688	1074342
# Attack samples	92034	556697	290089
Task	Multi-class	Multi-class	Binary
Unknown classes	CICDarknet2020	CICDarknet2020	CICDarknet2020

VI. EVALUATION

In this section, we demonstrate the effectiveness and efficiency of AEGIS-Net in addressing **CH1** and **CH2** by presenting experimental results and analyses under different experimental conditions.

A. Datasets

We use three benchmark datasets for network intrusion detection: Mal_TLS2023 [49], CICIDS2017, and CICIDS2018 [50]. In the CICIDS2017 dataset, we use all 14 types of malicious traffic samples. In the CICIDS2018 dataset, we select the traffic from the two days of 3.2 and 2.16 and choose three kinds of malicious traffic among them: DoS attacks, DoS attacks-SlowHTTPTest, and Bot, to form the attack samples, and the rest of the benign traffic as the normal samples. In the unknown traffic detection phase, we select the CICDarknet2020 [51] dataset as the unknown class traffic, which includes eight types of darknet traffic such as Audio-Stream, Browsing, Chat, Email, P2P, Transfer, Video-Stream, and VOIP. These darknet traffic types actually come from the Tor and VPN traffic in the ISCX-Tor2016 [52] and ISCXVPN2016 [53] datasets. As shown in Table III, these datasets provide sufficient benign traffic and a wide range of attack traffic categories, such as DDoS attacks, brute force, scanning attacks, etc. These four representative datasets allow us to evaluate the proposed method comprehensively.

B. Implementation Details

1) *Traffic Preprocessing:* For the Mal_TLS2023 dataset, we use the preprocessed dataset provided by the author. Each network flow in this dataset has a feature vector of 117 dimensions, which includes common statistical features as well as additional features specific to TLS traffic, such as cipher suite, cipher suite length, and certificate length. The dataset contains a total of 23 classes and 92,034 TLS malicious network flows. This dataset is class-balanced. The CICIDS2017 dataset contains 2,829,385 network flows, including 2,272,688 benign flows and 556,697 malicious flows across 14 classes, such as Bot, DDoS, DoS GoldenEye, SSH-Patator, and Infiltration. The CICIDS2018 contains 1,364,431 network flows, which include 1,074,342 benign flows with 290,089 malicious flows, which consist of three different kinds of malicious flows, which we uniformly consider as malicious flows in the experiments of this paper. Since these two datasets are class-imbalanced, particularly with

regard to benign traffic, we use an undersampling technique to reduce the number of samples from the majority classes. The CICDarknet2020 dataset contains 119,562 malicious traffic flows, covering diverse types of darknet traffic. These datasets were split into training and test sets at a 7:3 ratio.

2) *Noise Settings*: We have designed symmetric and asymmetric label noise types to test our model based on real-world scenarios. In the symmetric noise scenario, the sample labels are randomly shuffled to other classes with a fixed probability. In the asymmetric noise scenario, we simulate labeling errors of easily confused categories in real network traffic. For example, in the CICIDS2017 dataset, since infiltration attacks often use port scanning as a precursor, a fixed proportion of PortScan traffic is randomly labeled as Infiltration. DoS Slowhttptest and DoS slowloris are both variants of DoS attacks with similar traffic patterns, and a fixed proportion of DoS Slowhttptest traffic is randomly labeled as DoS slowloris.

3) *Training Details*: The network uses a deep residual neural network whose core architecture consists of a cascade of multiple residual blocks, each containing a double convolution operation. The four-level residual block constructs a feature pyramid through a progressive channel expansion strategy, which allows the network to gradually capture abstract features at the deeper level while retaining fine-grained detail information. The global adaptive average pooling layer compresses each channel response into a scalar and finally generates a 1024-dimensional discriminative feature vector.

All experiments are performed on an Ubuntu 20.04 system equipped with an Intel Xeon Gold 5218R CPU@2.10 GHz 80 GB RAM and a Tesla V100 32 GB GPU. Python 3.10.13 and Pytorch 2.1.1 are used to build our model. We use the SGD optimizer for model training, with an initial learning rate of 0.001, 50 epochs, 20 pre-training epochs, a batch size of 128, $\alpha = 0.1$, and $\beta = 0.5$. In the unknown malicious traffic detection phase, the KNN algorithm is set with $k = 50$.

4) *Baselines*: We have selected several state-of-the-art noise label handling methods and unknown malicious traffic detection methods to compare with AEGIS-Net in this paper:

i) *Noise Label Handling Methods*:

- *Co-teaching* [23] uses two neural networks, with the current network passing the samples it considers clean to the other network for training.
- *MCRE* [19] balances robust training and data cleaning, resulting in a cleaner dataset and a more accurate classification model.
- *RAPIER* [24] leverages the characteristic that normal traffic distribution is compact and malicious traffic distribution is dispersed. It estimates the distribution and corrects sample labels using an autoregressive generative model.

ii) *Unknown Malicious Traffic Detection Methods*:

- *RoFi* [38] enhances robustness to packet direction and timing by using a traffic aggregation matrix, providing a more robust representation of traffic than existing website fingerprinting defense strategies. RoFi outperforms existing WF attacks in two open-world scenarios.

- *Trident* [41] trains a one-class classifier for each known class, enabling it to handle both sample and class increments. It facilitates fine-grained unknown traffic detection and model incremental updates.
- *RFG-HELAD* [39] combines a DNN with contrastive learning to create a K-classification model and employs a distance-based out-of-distribution detection algorithm for identifying unknown attacks.

C. Known Attack Classification Experiments (To CHI)

Deep networks have powerful memory capabilities, allowing them to achieve high accuracy on the training set even in the presence of a high proportion of noisy labels by memorizing the random noise in the training data. However, the generalization performance on the test set significantly drops. Therefore, designing a noise label handling module is crucial in complex scenarios such as encrypted malicious traffic classification. For CHI, we set up different experimental environments to validate the effectiveness of the AEGIS-Net noise label correction module.

Tables IV, V, and VI show the performance of AEGIS-Net in classifying known class traffic across three public datasets. From these results, we can draw the following conclusions:

1) *AEGIS-Net has stronger robustness to noise*: In symmetric noise scenarios, under low noise conditions, AEGIS-Net maintains high classification accuracy, similar to comparative methods like MCRE and Co-teaching. For example, in the Mal_TLS2023 dataset with a symmetric noise rate of 0.1, AEGIS-Net achieves a classification accuracy of 96.69% and an F1-score of 96.37%. In high-noise scenarios, most methods, such as Co-teaching and RoFi, see a significant performance drop, with classification accuracy falling below 70%. However, AEGIS-Net still maintains a high level of performance. In asymmetric noise scenarios, asymmetric noise has a greater impact on models. In the Asym-50% scenario, Co-teaching's F1-score on the Mal_TLS2023 dataset drops to 64.98%, while AEGIS-Net's F1-score remains stable above 89%, indicating its better adaptability to complex noise.

2) *AEGIS-Net has the strongest cross-dataset performance consistency*: On datasets that focus more on malicious TLS traffic classification, AEGIS-Net's F1-score in the Asym-50% noise scenario significantly outperforms comparative methods, such as RAPIER's 83.56% versus AEGIS-Net's 89.87%, demonstrating its precise capture of encrypted traffic features. On the CICIDS2017/2018 dataset, which contains diverse attack types, methods like Trident, RoFi, and RFG-HELAD, which lack noise label processing modules, show large performance fluctuations under high noise. In contrast, AEGIS-Net's accuracy and F1-score decline only slightly, indicating its superior ability to generalize across heterogeneous attack patterns.

3) *AEGIS-Net demonstrates the best label noise handling capability*: Looking across each row of the tables, whether for multi-class or binary classification tasks, under various noise conditions, MCRE outperforms Co-teaching and RAPIER in label noise handling, as MCRE is a model compatible with both

TABLE IV
THE CLOSED-WORLD PERFORMANCE OF ALL METHODS UNDER DIFFERENT NOISE RATIOS ON THE MAL_TLS2023 DATASET

Methods	Sym-10%		Sym-30%		Sym-50%		Asym-10%		Asym-30%		Asym-50%	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Co-teaching	92.26	92.49	81.48	81.35	70.95	71.34	93.39	93.32	84.80	85.23	63.73	64.98
MCR	94.30	94.19	93.15	93.12	91.63	91.36	94.03	94.95	93.16	93.19	92.59	92.81
RAPIER	88.72	88.93	85.54	86.59	80.23	81.37	91.85	92.04	87.75	88.16	82.42	83.56
RoFi	89.88	90.21	79.58	80.41	69.27	69.08	90.03	90.67	81.58	81.69	62.98	63.87
Trident	91.23	91.48	77.48	77.16	64.12	65.49	92.47	92.73	79.52	79.28	66.31	66.54
RFG-HELAD	90.48	90.26	75.34	75.45	63.39	62.94	91.24	92.47	77.39	77.92	65.12	65.10
AEGIS-Net	96.69	96.37	96.53	96.30	95.84	95.45	96.87	96.64	96.35	96.00	89.02	89.87

"Sym" and "Asym" denote the symmetric and asymmetric label noise, respectively.

TABLE V
THE CLOSED-WORLD PERFORMANCE OF ALL METHODS UNDER DIFFERENT NOISE RATIOS ON THE CICIDS2017 DATASET

Methods	Sym-10%		Sym-30%		Sym-50%		Asym-10%		Asym-30%		Asym-50%	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Co-teaching	91.84	91.26	88.32	88.49	82.48	82.45	92.58	92.64	89.98	89.81	82.87	83.05
MCR	96.36	96.92	94.47	94.92	91.72	91.35	96.23	96.81	95.17	95.48	92.24	91.99
RAPIER	86.84	86.49	81.50	81.59	74.59	75.90	87.98	87.60	80.72	79.94	75.52	74.84
RoFi	88.37	88.12	73.56	73.44	65.26	65.57	90.58	90.75	70.41	71.36	68.78	68.45
Trident	90.53	90.35	75.38	75.64	67.28	67.62	91.63	91.63	69.61	68.95	50.65	51.56
RFG-HELAD	92.67	92.37	80.63	80.42	70.54	70.73	92.72	93.01	73.34	73.56	62.32	61.48
AEGIS-Net	97.69	97.74	96.62	96.88	92.76	92.46	98.00	98.19	95.97	95.62	81.32	81.64

TABLE VI
THE CLOSED-WORLD PERFORMANCE OF ALL METHODS UNDER DIFFERENT NOISE RATIOS ON THE CICIDS2018 DATASET

Methods	Sym-10%		Sym-30%		Sym-50%		Asym-10%		Asym-30%		Asym-50%	
	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Co-teaching	92.43	92.38	90.58	90.02	84.68	84.92	93.29	93.18	91.49	91.73	86.51	86.47
MCR	96.87	96.85	95.54	95.29	92.58	92.83	96.92	96.94	95.25	95.38	92.85	92.73
RAPIER	88.35	88.46	83.39	83.57	76.28	76.12	89.23	89.62	84.58	84.55	77.47	77.71
RoFi	89.45	90.67	74.68	74.86	67.36	68.68	91.47	91.46	72.65	72.43	65.45	65.87
Trident	91.34	91.46	76.57	76.83	69.43	69.34	92.55	92.72	73.51	73.46	62.41	62.45
RFG-HELAD	93.67	93.77	81.64	81.98	72.83	72.54	92.69	92.53	75.56	75.05	69.46	69.02
AEGIS-Net	99.80	99.93	99.77	99.97	99.70	99.91	99.78	99.91	99.76	99.92	99.73	99.91

data cleaning and robust training. However, as the noise rate increases, AEGIS-Net outperforms these three models. This is due to AEGIS-Net's excellent label noise handling capability, where it selects multiple prototypes for each class and accurately corrects noisy labels based on the distance of samples from the prototypes.

Chen et al. [54] proves the theoretical upper bound for the classification accuracy of models trained on datasets with label noise. For symmetric noise rate r , the upper bound for test accuracy is:

$$P(y^f = y) = (1 - r)^2 + \frac{r^2}{c - 1}, \quad (19)$$

where y^f denotes the predicted label by neural network f , and c is the number of classes. For asymmetric noise rate r , the upper bound for test accuracy is:

$$P(y^f = y) = (1 - r)^2 + r^2. \quad (20)$$

Table VII presents a comparison of AEGIS-Net's known class traffic classification performance and theoretical performance under different noise rates, as well as the proportion of pure data after label correction. From this, it can be observed that the theoretical accuracy of the classifier significantly decreases as the noise rate increases. For example, on the Mal_TLS2023 dataset, the theoretical accuracy is 26.14% when the symmetric noise rate is 50%. However, due to the presence of the label correction module, the model's actual performance far exceeds the theoretical value.

Data cleaning is a core method for improving model performance under noisy labels, and the quality of the cleaned data directly impacts the classification results. From the performance on the three datasets, data cleaning significantly reduced the actual training noise rate, resulting in a higher proportion of pure data, which helps mitigate the impact of label noise on model performance. In the absence of label noise, AEGIS-Net achieves classification accuracies of 96.86%, 98.23%, and 99.83% on

TABLE VII
THE PERFORMANCE OF AEGIS-NET IN CLASSIFYING KNOWN CLASSES OF TRAFFIC UNDER DIFFERENT NOISE SCENARIOS UNDER THREE DATASETS, WITH THE PERCENTAGE OF PURE DATA AFTER LABEL CORRECTION

Dataset	Noise rate	0%	Sym					Asym				
			10%	20%	30%	40%	50%	10%	20%	30%	40%	50%
Mal_TLS2023	Theoretically	100.0	81.05	64.18	49.41	36.73	26.14	82.00	68.00	58.00	52.00	50.00
	Accuracy	96.86	96.69 ($\uparrow 15.64$)	96.62 ($\uparrow 32.44$)	96.53 ($\uparrow 47.12$)	96.25 ($\uparrow 59.52$)	95.84 ($\uparrow 69.70$)	96.87 ($\uparrow 14.87$)	96.68 ($\uparrow 28.68$)	96.35 ($\uparrow 38.35$)	95.79 ($\uparrow 43.79$)	89.02 ($\uparrow 39.02$)
	Percentage	–	96.61	96.60	96.46	96.21	95.68	96.79	96.64	96.43	95.61	88.15
CICIDS2017	Theoretically	100.0	81.07	64.29	49.64	37.14	26.79	82.00	68.00	58.00	52.00	50.00
	Accuracy	98.23	97.69 ($\uparrow 16.62$)	97.04 ($\uparrow 32.75$)	96.62 ($\uparrow 46.98$)	95.08 ($\uparrow 57.94$)	92.76 ($\uparrow 65.97$)	98.00 ($\uparrow 16.00$)	97.66 ($\uparrow 29.66$)	95.97 ($\uparrow 37.97$)	84.88 ($\uparrow 32.88$)	81.32 ($\uparrow 31.32$)
	Percentage	–	97.70	97.01	96.85	95.18	92.89	98.10	97.56	96.08	84.93	81.12
CICIDS2018	Theoretically	100.0	82.00	68.00	58.00	52.00	50.00	82.00	68.00	58.00	52.00	50.00
	Accuracy	99.83	99.80 ($\uparrow 17.80$)	99.76 ($\uparrow 31.76$)	99.77 ($\uparrow 41.77$)	99.73 ($\uparrow 47.73$)	99.71 ($\uparrow 49.71$)	99.79 ($\uparrow 17.78$)	99.70 ($\uparrow 31.70$)	99.76 ($\uparrow 41.76$)	99.73 ($\uparrow 47.73$)	99.73 ($\uparrow 49.73$)
	Percentage	–	99.82	99.73	99.73	99.73	99.70	99.80	99.74	99.74	99.73	99.71

TABLE VIII
THE OPEN-WORLD PERFORMANCE OF ALL METHODS UNDER DIFFERENT NOISE RATIOS ON MAL_TLS2023, CICIDS2017, AND CICIDS2018 DATASET

Dataset	Methods	Sym-10%		Sym-30%		Sym-50%		Asym-10%		Asym-30%		Asym-50%	
		Acc	AUC	Acc	AUC	Acc	AUC	Acc	AUC	Acc	AUC	Acc	AUC
Mal_TLS2023	RoFi	85.82	86.42	82.23	84.39	74.19	76.92	86.78	87.92	79.28	82.54	75.39	77.29
	Trident	71.30	77.23	68.31	70.54	55.18	59.45	72.43	78.82	69.28	72.43	56.89	59.88
	RFG-HELAD	72.23	74.72	69.85	71.85	62.59	69.30	71.23	73.94	67.13	69.29	61.82	64.78
	AEGIS-Net	98.76	100.0	98.76	100.0	98.76	100.0	98.76	99.78	98.76	99.94	98.76	100.0
CICIDS2017	RoFi	83.34	89.43	79.74	82.73	70.12	74.44	82.49	87.22	73.45	79.35	54.56	61.53
	Trident	77.42	82.63	71.87	76.84	51.84	56.62	71.95	80.14	66.24	74.52	50.69	57.71
	RFG-HELAD	78.84	84.82	70.52	74.93	56.47	70.45	71.63	75.53	61.46	69.43	45.34	55.45
	AEGIS-Net	98.69	99.35	98.69	100.0	98.69	99.73	98.69	99.17	98.69	97.80	98.69	98.60
CICIDS2018	RoFi	86.31	89.62	80.57	84.48	76.58	80.45	84.83	90.66	76.36	79.94	67.73	71.43
	Trident	84.26	88.43	77.39	81.65	62.45	73.63	82.65	85.75	73.52	78.45	56.82	60.55
	RFG-HELAD	84.62	87.35	74.63	83.44	63.68	72.74	81.37	86.23	68.34	74.60	49.89	59.81
	AEGIS-Net	98.36	99.83	98.33	99.70	98.33	99.64	98.34	99.89	98.33	99.73	98.34	99.69

the three datasets, respectively. Even when the asymmetric noise rate reaches 50%, AEGIS-Net’s classification accuracies on the three datasets are still 89.02%, 81.32%, and 99.73%, showing improvements of 39.02%, 31.32%, and 49.73%, respectively, compared to the theoretical accuracy. This demonstrates that AEGIS-Net has excellent label noise correction capabilities.

D. Unknown Attack Detection Experiments (To CH2)

Noisy label samples form pseudo-class clusters in the feature space, occupying the open region that should belong to unknown attacks. This causes the unknown attack detection performance of NIDS to be inevitably affected by noisy labels. Therefore, designing a noise-resistant method for unknown attack detection is crucial. For CH2, we set the CICDarknet2020 dataset as the unknown class attack traffic and validate the effectiveness and

efficiency of AEGIS-Net for unknown attack traffic detection in three different experimental environments.

Table VIII shows a comparison of AEGIS-Net with other methods in the open world. The following conclusions can be drawn:

1) *AEGIS-Net exhibits significantly better noise robustness compared to baseline methods:* In both symmetric and asymmetric high-noise scenarios, AEGIS-Net’s detection accuracy and AUC values are close to or reach 100%. For example, on the Mal_TLS2023 dataset, AEGIS-Net achieves an accuracy of 98.76% under Sym-50% noise, while RoFi, Trident, and RFG-HELAD only achieve 74.19%, 55.18%, and 62.59%, respectively. This indicates that AEGIS-Net’s label correction mechanism effectively suppresses noise interference in the feature space, whereas traditional methods experience a significant drop in performance under noise.

2) *AEGIS-Net has a stronger ability to distinguish unknown attacks:* The AUC value reflects the model’s ability to differenti-

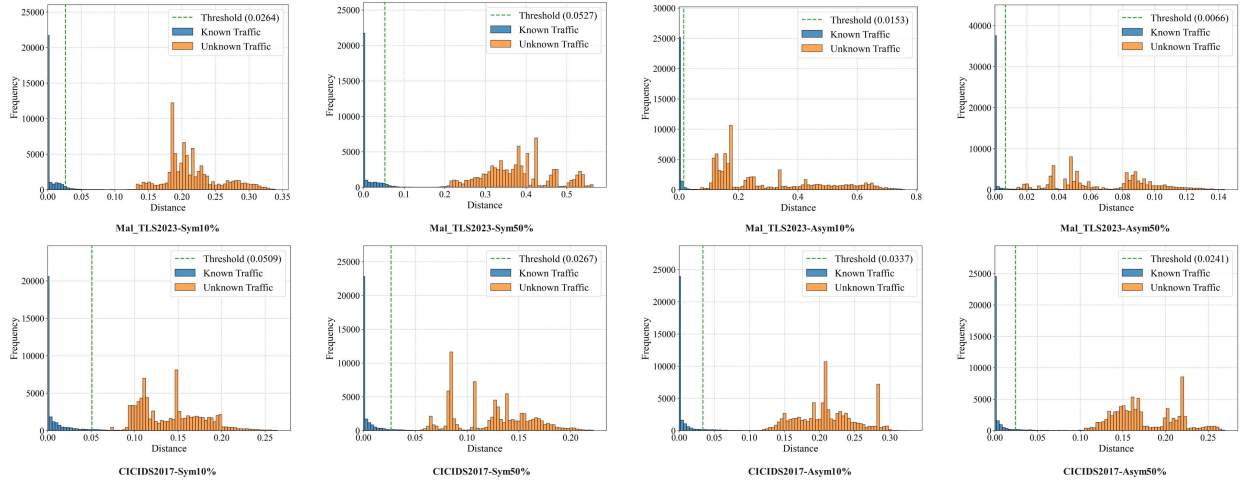


Fig. 4. The score distribution of known and unknown samples of AEGIS-Net on different datasets, with the green vertical line representing the threshold calculated based on the training set.

ate between known and unknown categories. AEGIS-Net’s AUC is higher than that of baseline methods in all scenarios, and it reaches 100% under multiple noise conditions. For instance, in the Asym-50% scenario of the CICIDS2017 dataset, AEGIS-Net achieves an AUC of 98.60%, while RFG-HELAD only achieves 55.45%. This demonstrates that AEGIS-Net constructs a compact known class feature distribution through multi-prototype contrastive learning, significantly improving the clarity of the unknown class boundary.

3) *AEGIS-Net shows excellent cross-dataset generalization:* In three heterogeneous datasets, the standard deviations of AEGIS-Net’s accuracy and AUC are less than 0.5% and 0.3%, respectively. In contrast, baseline methods like Trident show an accuracy fluctuation of 34.81% in the Sym-50% scenario of CICIDS2018, dropping from 84.26% to 62.45%. This indicates that AEGIS-Net’s model-agnostic detection paradigm is insensitive to traffic feature distribution differences, showing stronger adaptability to different scenarios.

From Table VIII, it can be seen that AEGIS-Net’s performance in detecting unknown malicious traffic is independent of the noise rate. This is because AEGIS-Net’s detection model is non-parametric, and the detection module does not participate in model training. Although the model’s closed-world performance is strongly correlated with different noise rates, the detection of unknown traffic remains unaffected. This is actually desirable. If open-world detection performance depended on closed-world detection performance, the model’s closed-world performance would inevitably degrade as the noise rate increases, leading to a deterioration in open-world detection performance. However, in this study, although AEGIS-Net’s closed-world performance slightly decreases as the noise rate increases, it has little effect on open-world detection performance.

Fig. 4 shows the anomaly scores for different samples, where the green line represents the classification threshold determined based on the known class samples. Samples exceeding this threshold are classified as unknown malicious traffic. From this, it can be observed that, regardless of the noise rate or the dataset used, there is almost no overlap between the scores of

known and unknown class samples. Furthermore, almost 100% of the anomaly scores for known class samples are below 0.1, while the anomaly scores for unknown class traffic are widely distributed. For example, in the Mal_TLS2023-Asym10% scenario, the anomaly scores for unknown class traffic range from [0.05, 0.8], in sharp contrast to the scores of known class samples. Nearly 100% of the unknown class samples have anomaly scores greater than the threshold, enabling a clear distinction between known and unknown class traffic. This explains the excellent performance of AEGIS-Net as shown in Table VIII.

As shown in Fig. 5, from the visualization results of the penultimate layer features of different datasets, it can be observed that the penultimate layer features from different datasets exhibit a significant multi-cluster structure in the embedding space. This phenomenon directly validates the rationality of selecting multiple prototypes for each class.

For example, in the CICIDS2018 dataset, although the training set contains only two major categories, benign and malicious traffic, benign traffic includes various network behaviors such as normal web browsing, sending and receiving emails, and file transfers. This can be corroborated by the visualization results showing multiple sub-clusters within the benign traffic. If only a single prototype were used, it would fail to capture the intra-class diversity, leading to blurred feature representations. In contrast, multiple prototypes can accurately represent the intra-class structure. In the CICIDS2017 dataset, focusing specifically on DDoS attacks, there are various attack types, such as Flood Attacks, Amplification Attacks, Layer 7 Attacks, and Hybrid Attacks, each with significant differences in traffic patterns, such as packet frequency, protocol types, and load distribution. This results in a clear separation of sub-clusters in the feature space. A single prototype cannot distinguish between attack types, whereas multiple prototypes can model different attack patterns separately, improving classification and detection accuracy.

Furthermore, the high separation between different classes indicates that the model has learned discriminative features

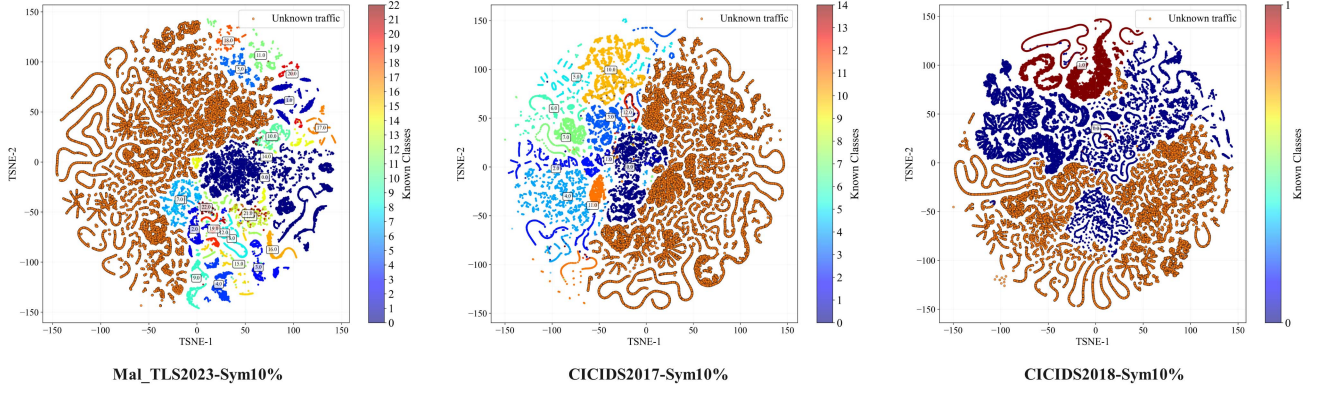


Fig. 5. Penultimate layer feature space visualization, where orange dots indicate unknown malicious traffic.

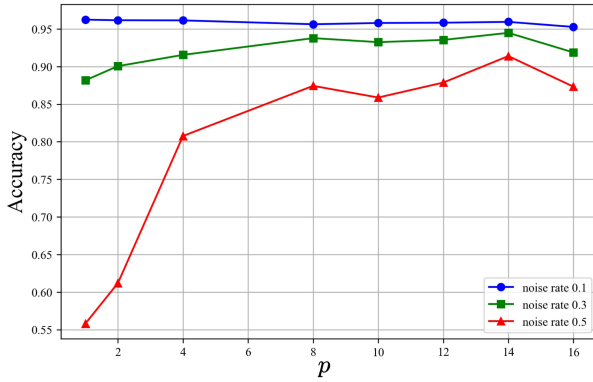


Fig. 6. In the Mal_TLS2023 dataset, the impact of the number of prototypes p on the closed-world classification performance of AEGIS-Net under symmetric noise scenarios.

to distinguish between various attack types. The contrastive loss further strengthens this separation, making the decision boundaries clearer. The orange scatter in the figure indicates the location of unknown malicious traffic, indicating that unknown malicious traffic exhibits high randomness and diversity in the feature space. These orange scatter overlap with other colored points (representing known traffic) but mostly lie on the boundary of known traffic, suggesting that darknet traffic shares some feature similarities with known traffic. However, darknet traffic has unique encryption methods, communication patterns, or load characteristics, causing it to be separated from known traffic in the feature space.

E. Sensitivity Analysis

In this subsection, we perform sensitivity analysis on the critical hyperparameter in AEGIS-Net.

1) *Effect of p* : From Fig. 6, it can be observed that when the noise level is low, the model's classification performance is not significantly affected by different values of p , with the classification accuracy remaining around 95%. Under low noise, the feature distributions of samples within the same class are concentrated, and a small number of prototypes can cover the intra-class diversity. Increasing the number of prototypes (p)

does not notably improve classification performance and may even introduce redundancy. Therefore, increasing the number of prototypes is not necessary to enhance performance. In the case of high noise, if each class only has one prototype, the model's classification accuracy drops to 56%. As the value of p increases, the model's classification accuracy gradually improves. However, when p exceeds 16, the model's accuracy begins to decline. Under 50% noise, a single prototype may be severely contaminated by noisy samples, such as label errors causing prototype shifts, which prevent it from effectively representing the true class center. In low-noise scenarios, a small number of prototypes can adequately represent the class without redundancy. In high noise scenarios, more prototypes are required to cover intra-class diversity and correct for noise. Therefore, in our experiment, we choose $p = 14$ to balance robustness across different noise levels.

2) *Effect of k* : Table IX shows the impact of the hyperparameter k on the performance of unknown malicious traffic detection and the corresponding detection time. When the value of k is small, ranging from 10 to 1000, the detection accuracy remains high, indicating that smaller k values effectively capture the local features of the data, making them suitable for detecting unknown malicious traffic. However, when k exceeds 2000, the accuracy drops sharply to about 23.59% due to underfitting, as more neighbors may include samples from different categories, thus affecting the classification results. In extreme cases, such as when k equals the number of training samples, the output classification will be the most frequent class, rendering the distance irrelevant. Moreover, as the value of k increases, the average detection time per flow also increases, but the overall difference is not significant. When $k = 50$, the average time spent on each flow in the KNN algorithm is 0.0353 ms. Considering both the detection accuracy for unknown classes and the time cost, we choose $k = 50$ in this paper.

3) *Effect of m* : As shown in Table X, the closed-world classification accuracy is not significantly affected by different sample sizes m . This is because m is used to reduce the computational and storage overhead of the similarity matrix S , which has a size of $n \times n$, where n is the number of samples in the training set. When n is large, computing and storing S becomes time-consuming and expensive. By randomly sampling

TABLE IX

IN THE MAL_TLS2023 DATASET, THE EFFECT OF THE PARAMETER k IN THE KNN ALGORITHM ON THE DETECTION PERFORMANCE OF UNKNOWN MALICIOUS TRAFFIC UNDER SYMMETRIC NOISE OF 0.1

k	10	20	50	100	200	500	1000	2000	3000	5000
Unknown Attack Detection Acc(%)	98.76	98.76	98.76	98.76	96.89	96.89	98.76	23.59	23.59	23.60
Time(ms)	0.0345	0.0351	0.0353	0.0355	0.0357	0.0371	0.0457	0.0432	0.0463	0.0544

In the experiment, we use the Faiss library for efficient nearest neighbor search, with 80 threads to fully utilize the hardware performance. The search space consists of 64,423 training samples, and the number of samples to be detected is (27,611 + 83,693) samples.

TABLE X

IN THE MAL_TLS2023 DATASET, THE EFFECT OF THE SAMPLE SIZE m ON THE CLASSIFICATION OF KNOWN CLASS TRAFFIC UNDER SYMMETRIC NOISE OF 0.1

m	320	640	960	1280	1600	2240
Acc(%)	95.31	94.09	95.87	96.69	94.86	94.80

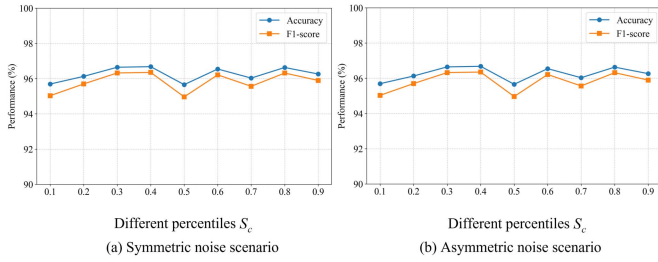


Fig. 7. The impact of the percentile S_c on model performance in the Mal_TLS2023 dataset with a noise rate of 0.1.

m samples (where $m < n$), the size of the similarity matrix is reduced to $m \times m$, significantly reducing the complexity of computation and storage. Although the sample size m reduces the number of samples, the randomly sampled samples still effectively represent the distribution of the entire training set. Therefore, the similarity matrix S computed based on these sampled samples can still reflect the similarity relationships between samples in the training set. In this paper, we choose $m = 1280$ for the experiments.

4) *Effect of S_c* : As shown in Fig. 7, within the range of percentiles S_c from 0.1 to 0.9, the performance fluctuation of AEGIS-Net under both symmetric and asymmetric noise scenarios is less than $\pm 1.5\%$. This stability arises from the core property of the density calculation ρ_i in (6): when S_c varies within the effective range, only the absolute density values of the samples are adjusted, while the relative density ranking between samples remains highly consistent. Therefore, the prototype selection remains unchanged, and the final performance is not significantly affected.

5) *Effect of α, β* : The weight factors $\alpha, \beta \in [0, 1]$ are used to balance the loss function in (5). We perform hyperparameter tuning within the range of 0.1 to 0.9. Under the condition of symmetric noise rate of 0.1 on the Mal_TLS2023 dataset, the classification accuracy is shown in Fig. 8. The optimal performance of AEGIS-Net is achieved when $\alpha = 0.1$ and $\beta = 0.5$, reaching 96.69%. Therefore, we set $\alpha = 0.1$ and $\beta = 0.5$ as the final values.

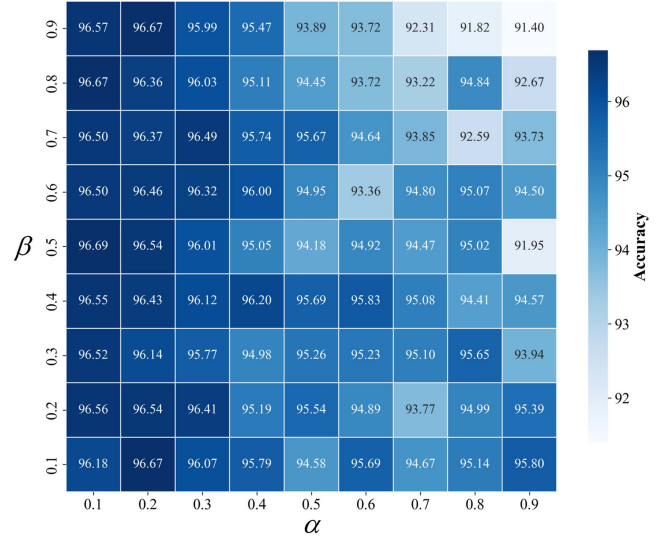


Fig. 8. Accuracy heatmap for hyperparameters α and β of AEGIS-Net on Mal_TLS2023 dataset with a symmetric noise rate of 0.1.

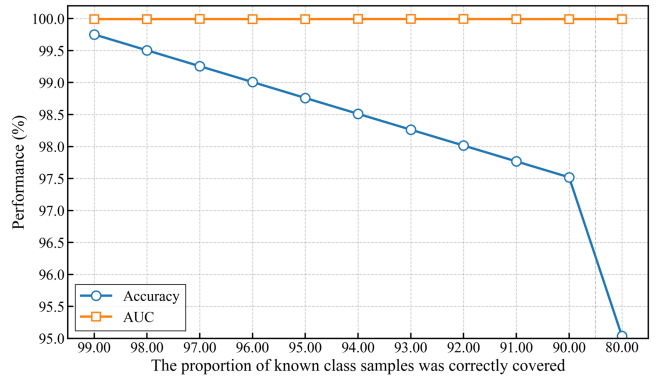


Fig. 9. Effect of different thresholds λ on AEGIS-Net performance on the Mal_TLS2023 dataset with a symmetric noise rate of 0.1.

6) *Effect of Different Threshold λ* : As shown in Fig. 9, the AUC remains stable across all values of λ , demonstrating that the model's ability to capture unknown threats is completely unaffected by the threshold. Although lowering the threshold gradually improves performance, with the detection accuracy increasing from 98.759% to 99.752%, we believe that in practical applications, ensuring a 95% pass rate for known class traffic and achieving an accuracy of 98.759% is sufficient. The

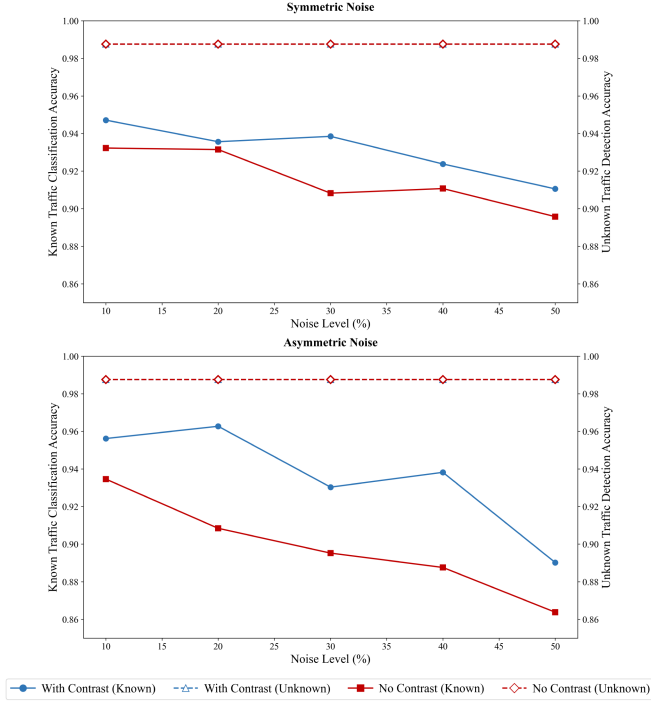


Fig. 10. The impact of contrastive loss on model performance in the Mal_TLS2023 dataset.

1% performance improvement from fine-tuning the threshold seems negligible.

F. Ablation Study

1) *Effect of Contrastive Loss*: The top and bottom figures in Fig. 10 illustrate the impact of contrastive loss on AEGIS-Net under symmetric and asymmetric noise scenarios, respectively. It can be observed that, regardless of the presence of contrastive loss, the detection accuracy for unknown malicious traffic remains identical under both noise conditions, at 98.76%. This indicates that the detection of unknown malicious traffic is not significantly affected by different noise conditions. Contrastive loss, however, significantly improves the robustness of classification for known classes, especially under asymmetric noise conditions, where the performance gap expands as the noise level increases. For instance, at a 40% noise level, the gap reaches 5.06%.

Specifically, while the contrastive loss optimizes the feature space to improve the compactness of the known classes, the unknown class samples naturally move away from the known class clusters due to distribution differences. As a result, the detection performance mainly depends on the geometric structure of the feature space rather than the classifier parameters, so there is no negative impact on the detection of unknown classes. Contrastive loss, based on the relationships between samples (positive/negative pairs) rather than individual labels, reduces direct reliance on incorrect labels. Optimizing the intra-class and inter-class relationships in the feature space significantly

TABLE XI
ABLATION STUDY OF AEGIS-NET ON MAL_TLS2023 DATASET WITH A SYMMETRIC NOISE RATE OF 0.1, WHERE ACCURACY AND F1-SCORE REPRESENT THE DETECTION METRICS FOR KNOWN TRAFFIC

Method	Sym-10%		Sym-30%		Sym-50%	
	Acc	F1	Acc	F1	Acc	F1
AEGIS-Net	96.69	96.37	96.53	96.30	95.84	95.45
w/o Pretraining	91.41	89.60	77.22	72.87	37.22	20.19
w/o Label Correction	94.82	94.89	92.71	92.82	91.13	91.49
w/o Encoder E	84.02	82.36	81.24	77.17	80.67	79.46

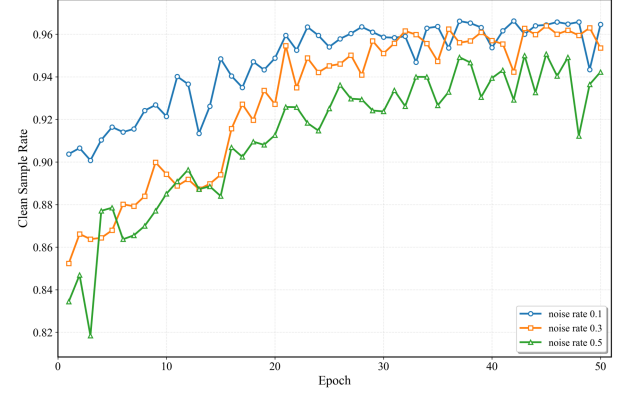


Fig. 11. In the Mal_TLS2023 dataset, clean sample rate over training epochs (Symmetric Noise). The pretraining phase consists of the first 20 epochs.

enhances the robustness of classification for known classes under noisy labels.

2) *Effect of Pretraining and Label Correction*: As shown in Table XI, we investigate the effectiveness of the pretraining procedure and the label correction module proposed in this paper. Without pretraining, the model's performance drops significantly, and under the Sym-50% condition, the model almost completely fails. This indicates that pretraining the model before the label correction module begins to operate is crucial, as it helps the encoder E learn good representational capabilities. After removing encoder E , the model's performance shows a significant decline in different noise environments. Specifically, in the 10% symmetric noise environment, the classification accuracy drops from 96.69% to 84.02%, a decrease of 12.67%; in the 50% symmetric noise environment, the accuracy falls from 95.84% to 80.67%, a decrease of 15.17%. The F1 score exhibits a similar trend, indicating a noticeable reduction in the overall discriminative ability of the model. This outcome demonstrates that encoder E , as the core component of feature extraction, plays a crucial role in the model's robustness under noisy conditions. Its effective representation capability is an essential foundation for AEGIS-Net to maintain high performance.

In the absence of the label correction module, a convolutional neural network trained using only cross-entropy and contrastive loss, while having some classification ability, performs poorly and falls far short of practical usability. As shown in Fig. 11, during the first 20 epochs, although the label correction module is not active, the neural network still exhibits some label

TABLE XII
AVERAGE TESTING TIMES OF DIFFERENT METHODS

Method	FET (ms)	CPT (ms)	Total time (ms)
RoFi	0.11	0.52	0.63
Trident	0.15	1.03	1.18
RFG-HELAD	0.16	0.41	0.57
AEGIS-Net	0.16	0.04	0.20

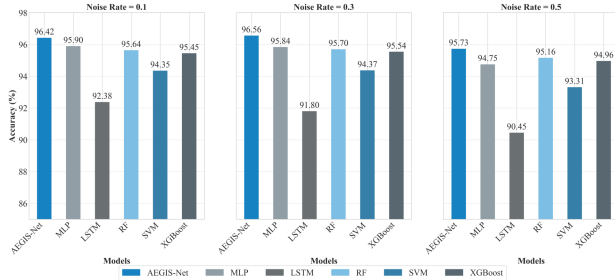


Fig. 12. Performance evaluation of various models under symmetric noise with label-corrected Mal_TLS2023 dataset.

correction capability, leading to a gradual increase in the clean sample ratio in the training set. After 20 epochs, when the label correction module becomes active, the clean sample ratio in the training set improves further. Under the Symmetric noise of 0.5, the final clean sample ratio reaches 94%, indicating that the proposed label correction module is effective.

G. Evaluation on Time Cost

We divide the complexity of the model into two aspects: feature extraction time and classifier prediction time [55], [56], [57]. FET stands for Feature Extraction Time, which refers to the time required for converting the raw traffic into a feature vector. The prediction time refers to the time spent in detecting each flow, while the total time is the sum of feature extraction time and classifier prediction time.

Table XII presents the feature extraction time and the average detection time per network flow for different methods. AEGIS-Net requires an average of 0.20 ms per network flow, which is better than RFG-HELAD's 1.18 ms, achieves the fastest speed. Trident maintains multiple single-class one-class learners, which results in slightly lower detection efficiency. Overall, AEGIS-Net does not have a significant disadvantage in detection efficiency compared to the current methods.

H. Scalability Analysis

Based on the corrected training set from AEGIS-Net, we trained and evaluated six models, including AEGIS-Net (the CNN model), MLP, LSTM, and three machine learning models (RF, SVM, and XGBoost). For the three deep neural networks—AEGIS-Net, MLP, and LSTM—we adopted a training strategy that combines cross-entropy loss with InfoNCE contrastive loss. As shown in Fig. 12, under different noise rates, the proposed AEGIS-Net model demonstrates the best performance on the

corrected training set. MLP, RF, and XGBoost also exhibit good classification performance, while the LSTM model performs relatively weaker. The results indicate that the proposed label correction module not only effectively improves model performance but also possesses strong versatility, serving as a plug-and-play preprocessing component that can be flexibly integrated with other models and play a positive role.

The unknown traffic detection module employs a model-agnostic K-nearest neighbor paradigm, which provides it with high versatility and integrability. Specifically, this module relies solely on the normalized feature vector of the input sample and a threshold λ that is adaptively determined based on the distance distribution of known class samples. Due to its non-parametric nature, this module does not depend on any specific classification architecture or parameter training phase; it makes decisions based directly on relative distances within the feature space. As a result, the module can function as an independent component that seamlessly integrates with different NIDS through a standardized interface, significantly enhancing the system's ability to detect unknown threats without affecting the existing detection processes.

VII. LIMITATIONS AND FUTURE WORKS

Our work has a few limitations. First, when the features of benign traffic change over time, our method may fail. In our improvement plan, detecting concept drift in traffic samples should be considered. Second, when facing adversarial traffic samples with incorrect labels, these samples are closer in distribution to the incorrect class, which could interfere with the noise label correction process. Future work will explore customized solutions, which will be a promising research direction. Lastly, as part of future work, we will investigate which components can run in parallel to maximize efficiency.

VIII. CONCLUSION

In the dynamic network environment, network intrusion detection systems face the dual challenges of label noise and unknown attack detection. This paper proposes a novel anti-noise detection framework, AEGIS-Net. AEGIS-Net dynamically selects multiple prototypes for each class through a multi-prototype strategy to achieve accurate noise label correction. It also designs a model-agnostic k-nearest neighbors distance detection paradigm, which determines unknown attacks by calculating the relative distance ratio between the test sample and known classes, eliminating the dependence on traffic distribution assumptions common in traditional methods. This makes the model more flexible and versatile. We prove, through the feature space stability hypothesis, that the detection threshold is independent of the noise rate, mathematically verifying the anti-noise property of unknown attack detection. Additionally, a series of experiments demonstrate the effectiveness of AEGIS-Net in known attack classification, unknown attack detection, and cross-scenario generalization ability. Moreover, we show performance improvement compared to the state-of-the-art baseline methods.

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